Muhammad Ahmad

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Summary

Machine Learning Engineer and Full-Stack Developer with hands-on experience building end-to-end, AI-powered web applications. Skilled in designing Retrieval-Augmented Generation (RAG) systems, automating data pipelines, and integrating scalable ML models with modern web stacks (MERN). Proven track record of shipping production-grade, user-centric applications from research to deployment.

Technical Skills

Languages: C/C++, Python, JavaScript, MySQL, PostgreSQL, PL/SQL, HTML/CSS
Frameworks & Libraries: React, Node.js, Express.js, FastAPI, Pandas, NumPy, Scikit-Learn, TensorFlow, LangChain
ML/AI & Data: RAG Pipelines, FAISS, pgvector, Sentence Transformers, LLM Integration (Groq, Llama), Feature Engineering
Tools & Platforms: Git, GitHub Actions, Supabase, MongoDB, Vercel, VS Code, Visual Studio, PyCharm, IntelliJ, Eclipse

Experience

CyberSecurity Zone Islamabad, Pakistan
Cybersecurity Intern *June 2025 – August 2025*

- Developed a secure file and disk encryption system in Python, addressing the risk of data loss and unauthorized access in real-world environments.
- Designed key management and recovery workflows using SMTP-based delivery, balancing encryption strength with usability and recoverability.
- Researched the impact of quantum computing on modern cryptographic systems and evaluated AI-driven defense strategies for future security infrastructure.

Rumi Lab Islamabad, Pakistan
Research Intern – Data Acquisition, Preprocessing & Model Training *May 2025 – Present*

- Built data acquisition and preprocessing pipelines to transform noisy, raw EEG signals into model-ready inputs.
- Engineered structured workflows for signal cleaning and feature extraction, improving downstream model performance.
- Contributed to predictive models for cognitive load analysis, applying machine learning to enhance educational systems through data-driven insights.
- Designed a system mapping student attention patterns to lecture design improvements, supporting AI-driven, human-centered adaptive learning.

Projects

Omnius Awakened – Dune AI Agent | *Python, FastAPI, Supabase, LiveKit, Groq LLM, React, Vercel* Aug 2026

- Built a context-aware real-time voice assistant (Omnius) to enable immersive, lore-accurate audio conversations for users exploring the *Dune* universe.
- Designed a Retrieval-Augmented Generation (RAG) pipeline using Supabase and Hugging Face embeddings to ground responses in relevant textual data.
- Leveraged LiveKit WebRTC, Groq Whisper STT, and Groq Orpheus TTS to achieve low-latency voice interactions, demonstrating real-time conversational systems in production.
- Engineered persona-driven system prompts to mimic a superior machine consciousness, personalizing outputs and ensuring narrative consistency.
- **Live:** omnius-awakened.vercel.app | **Repo:** github.com/muahmad232/dune-bot

Research Paper Screening Agent | *React, FastAPI, LangChain, Supabase, pgvector, Groq LLM, GitHub Actions* Jun 2026

- Built an autonomous AI agent that discovers, screens, analyzes, and recommends research papers tailored to each user's research profile, addressing information overload in academic research.
- Orchestrated a multi-stage agent pipeline with LangChain to generate search queries, retrieve papers from multiple scholarly sources, score relevance, perform LLM analysis, and identify research gaps.
- Implemented semantic similarity search using Supabase with pgvector, enabling retrieval based on research intent rather than exact keyword matching.
- Combined embedding similarity, LLM reasoning, keyword matching, and publication recency into a multi-signal ranking system, routing uncertain cases for human review.
- Automated daily execution with GitHub Actions and deployed a full-stack application with React, FastAPI, and Groq GPT OSS.
- **Live:** paper-agent-one.vercel.app | **Repo:** github.com/muahmad232/Research-Paper-AI

Football AI Predictor | *Python, Pandas, Scikit-Learn, React, Node.js* Dec 2025

- Built a predictive system trained on real-world EA Sports FC 25 data to evaluate and project player performance.
- Applied data preprocessing and feature engineering with Pandas to transform raw statistics into meaningful performance indicators.
- Implemented machine learning models in Scikit-Learn to predict player ratings and career trajectories, supporting decision-making in sports analytics.
- Developed a full-stack interface using React and Node.js to make model predictions accessible to end users.
- **Live:** football-player-creator.vercel.app | **Repo:** github.com/muahmad232/football-predictor

CyberLib | *React, Node.js, Express.js, MongoDB, Tailwind CSS* April 2025

- Built a centralized book discovery platform with personalized recommendations and conversational interaction, replacing scattered and impersonal alternatives.
- Integrated external APIs (Google Books, Open Library) to aggregate diverse datasets, demonstrating multi-source data integration.
- Implemented secure authentication and user-specific data storage using MongoDB, reflecting scalable backend design for personalized platforms.
- Designed a responsive UI using Tailwind CSS, emphasizing accessibility and user engagement.
- **Live:** cyber-lib.vercel.app | **Repo:** github.com/muahmad232/CyberLib

Education

National University of Sciences and Technology Islamabad, Pakistan
Bachelor of Science in Computer Science *Sept. 2023 – June 2027*

Courses and Certifications

- **Deep Learning Specialization** – DeepLearning.AI (Coursera)
- **Machine Learning Specialization** – DeepLearning.AI (Coursera)
- **Applied Data Science Lab** – WorldQuant University